

Individual Reflective Piece

I have experienced somewhat of an emotional rollercoaster in terms of my feelings towards this module and my motivation for undertaking tasks. Within this reflective piece I will look back at key learning points and barriers that I had to overcome to achieve the learning outcomes. In order to aid my reflection, I have employed Rolfe, Freshwater & Jasper's (2001) 'What?', 'So what?', 'What next?' format to structure my insights.

What?

I began this module with very little confidence, considering network security to be 'out of my comfort zone' seeing as I had very little experience in the field. This was compounded by the fact that my fellow students seemed to have backgrounds in network security:

Text from first seminar live chat:

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00:42:05      Femi O: A very good afternoon everyone, Unfortunately, I am not able to contribute to the seminar, currently at work
00:42:13      Femi O: Many thanks
01:02:18      Femi O: My name is Femi, an information Security Office with over nine years of working experience in the cyber security field specialising in governance risk and compliance (GRC), before transitioning to cybersecurity, I have worked as an assistant data privacy officer and Legal caseworker. the purpose of studying this course is to consolidate my knowledge furthermore.
By the way I resides in London.
01:05:18      Beran Necat: Thank you Femi
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Excerpts from first seminar transcript:

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127
00:11:47.290 --> 00:11:51.270
Franz Stephan: and I lived in the Uk for 7 years in Hampshire, in the past.

128
00:11:51.660 --> 00:11:55.529
Franz Stephan: where I had my own cybersecurity firm for a while.

129
00:11:55.750 --> 00:12:03.240
Franz Stephan: But at the time I only did vulnerability testing. Basically, that's how my managers, vulnerability testing creating reports for companies.
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164
00:14:05.120 --> 00:14:34.870
Deniesha Lloyd: Hi! I'm Denisha. Most people call me Nisha for short. I'm currently on my lunch break. So if, after I might have to drop off a little bit later. But I work for Hsbc. I'm a personal banker. I've been here for about 6 years now. I got into cyber security when I was pregnant in 2019, I've done a short course, and I find it really interesting. And now I that's why I decided to take up this course.
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184
00:15:46.940 --> 00:15:54.420
OmarOmar: Hi my name is Omar. I'm an architect mainly specializing in cloud security solutions

185
00:15:55.230 --> 00:16:00.050
OmarOmar: with 12 years experience from different areas

186
00:16:01.128 --> 00:16:07.770
OmarOmar: developing support infrastructure networking and security

187
00:16:08.494 --> 00:16:15.760
OmarOmar: currently residing in London. But I'm from Zanzibar Islands in a small island in East Africa.

188
00:16:16.430 --> 00:16:20.399
OmarOmar: married with 2 kids, 10 in full.

189
00:16:21.350 --> 00:16:29.370
OmarOmar: I have a my background, my educational background. I hold the bachelor degrees in degree in computer networks and security.

190
00:16:30.901 --> 00:16:34.308
OmarOmar: I hold several industrial qualifications.

191

214
00:18:53.520 --> 00:19:04.729
Abdulrahman Al Hammadi: my, my studies and my career have been going side by side. So I start my studies. I took my H and D in Britain

215
00:19:04.800 --> 00:19:25.470
Abdulrahman Al Hammadi: from Chichester University. It was about software, engineering and system development. Then I took my computer engineering from Abu Dhabi University. And after that, for about 4 years I've been attending intensive science courses

216
00:19:25.550 --> 00:19:34.559
Abdulrahman Al Hammadi: was focusing on the forensic side and cyber security side the same time. So it was like red team blue teams, whatever involved in between.

217
00:19:35.150 --> 00:19:38.539
Abdulrahman Al Hammadi: But I've been focusing mainly on the forensics.

218
00:19:38.650 --> 00:20:06.800
Abdulrahman Al Hammadi: So I have plenty of practices along with other departments in my country, also with departments in UK. We had couple of meetings in RSA. And with other firms, big companies like image services. They are providing forensics, tools, and they have a lot of partners that are showing, giving real scenarios and discussing main issues regarding to incidents.

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I found I had an initial reluctance to participate in collaborative discussions, reflecting Xie & Ke's (2011) findings that feelings of connection towards peers is the most significant influential factor on a students' subsequent motivations to contribute to collaborative discussions in which ideas are built upon. An additional barrier was the fact that this was my first module not to feature programming – something that I see as a strength of mine and an endeavour that I had taken most pleasure in during my 3 previous modules. These motivational barriers align with Shapiro et al.'s (2017) findings that knowledge and personal interest are among the most influential factors on student motivation.

So what?

I decided that I had to keep an open mind and seek to gain knowledge and interest in the module content in order to succeed. Fortunately, the first collaborative discussion topic – that of digital transformation and its security implications – was a matter close to my heart, working as the sole IT technician in an independent special school that is looking to introduce more and more digital processes:

Initial post

by [Todd Edge](#) - Saturday, 1 February 2025, 7:35 AM

Number of replies: 5

A fully digital enterprise can be defined as being a company that fulfils 100% of Doroiman & Sîrghi's (2024) indicators within their Digital Economy and Society Index (DESI); DESI was used to measure the digital performance of companies in Doroiman & Sîrghi's (2024) empirical study of how digital transformation of enterprises impacts economic growth. However, I would also argue that, to be a 'fully digital enterprise', the level of basic digital literacy amongst employees at an SME – a prerequisite for making effective use of digital technology (Doroiman & Sîrghi, 2024) – would also have to be at 100%.

As the human factor poses the largest risk to digital systems (Fairburn et al., 2021; Pillai, 2017), it could be argued that one of the biggest challenges for a fully digital small/medium enterprise (SME) would be ensuring continuous updating of employees' digital skills and cyber security practices to ensure that this risk was mitigated (Doroiman & Sîrghi, 2024). From personal experience, a consequence of a lack of digital literacy can affect business efficiency when new security procedures are brought in; at my place of work, many colleagues have struggled to adapt to using multi-factor authentication. Despite being sent a guide on how to use this method of signing in to their account, many were locked out of their account while they waited for me to show them, in-person, how to use said method.

There would be a number of challenges for a bricks and mortar SME wanting to become a digital enterprise. One of these challenges would be ensuring that employees' digital literacy is developed to an acceptable level (Doroiman & Sîrghi, 2024); this would be more of a challenge in areas such as Romania and Bulgaria, where only around 30% of the population have basic digital literacy (Doroiman & Sîrghi, 2024). Another challenge would be accessing financial resources to enable investment in new digital technologies and training (Doroiman & Sîrghi, 2024). Additionally, early stage adoption of new technologies can often see initial inefficiencies of operation and this can have a large negative financial impact on a company (Doroiman & Sîrghi, 2024).

Learning Outcome:

1. Identify and analyse security threats and vulnerabilities in network systems and determine appropriate methodologies, tools and techniques to manage and/or solve them.

Learning Outcome:

4. Articulate the legal, social, ethical and professional issues faced by information security and risk professionals.

Feedback from my tutor on my initial post was positive, further increasing my confidence in engaging with this module, while the suggestions for further research piqued my interest and motivated me to look further into this issue:



Re: Initial post

by Beran Necat - Saturday, 1 February 2025, 6:56 PM

Hi Todd and Shaikha,

I enjoyed reading your thoughts on this topic.

One area for further exploration is the impact of digital transformation on employees and company culture.

Discussing strategies for managing this transition smoothly would be valuable. Including examples of successful digital transformations could provide a more balanced perspective, offering insights and inspiration.

Emphasizing the role of data-driven decision-making and enhancing customer experience through digital tools would further enrich your discussions. These aspects are crucial for operational efficiency and competitiveness.

Any thought on this?

Regards, Beran

Xie & Ke (2011) suggest that tutors should seek to boost student confidence in their own abilities in order to boost student motivation to collaborate in online learning discussions, and I believe that my tutor achieved this through such feedback. Indeed, I was suitably motivated to undertake the suggested research before constructing a summary post based on my tutor's lines of enquiry:

Summary post

by [Todd Edge](#) - Friday, 7 March 2025, 4:24 PM

Number of replies: 0

To further consider which factors most influence a smooth transition into fully digital processes, I will use the findings from 2 studies that investigate successful digital transformations within the Malaysian manufacturing industry (Ahmed et al., 2024) and within the banking industry in Indonesia (Annisa, Siahaan & Lumbanraja, 2024), respectively.

Developing a digital culture is a significant first stage of digital transition and involves the integration of digital technology into all aspects of company processes – from communication to operations (Annisa, Siahaan & Lumbanraja, 2024; Ahmed et al., 2024). Fostering a positive digital organisational culture has been found to have a significant impact on improved employee digital readiness by means of improving digital efficiency and increased innovation (Ahmed et al., 2024).

To complement the creation of a digital culture, continuous improvement - made possible through tailored training - is paramount (Ahmed et al., 2024; Annisa, Siahaan & Lumbanraja, 2024); here, support and innovation are the most influential factors on effective digital transformation (Ahmed et al., 2024; Annisa, Siahaan & Lumbanraja, 2024). Indeed, employees given the motivation to innovate through the utilisation of digital tools were found to be much more likely to embed the use of said tools into their daily practices (Ahmed et al., 2024).

A third key factor in the pursuit of full digital adoption is effective leadership that oversees the management of digital technologies, guiding employees to effectively utilise specific tools to achieve organisational targets (Annisa, Siahaan & Lumbanraja, 2024). Both studies agree that resistance to technological adoption can have a negative impact on smooth digital transitions (Annisa, Siahaan & Lumbanraja, 2024; Ahmed et al., 2024). In investigating how such resistance can be mitigated, Ahmed et al. (2024) suggest the adoption of effective change management that sees transparent communication to employees of plans and implications of digital transformations.

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I also found that questions that I received in a peer response to my initial post were surprisingly insightful, delving deeper into the discussed issues and posing

interesting questions that inspired my further research into motivations behind employee adoption of security processes:

In reply to Todd Edge

PEER RESPONSE

by [Franz Stephan](#) - Friday, 7 February 2025, 10:37 AM

Hey Todd,

Your comment of fully digital enterprises and the challenges SMEs face in digital transformation is very self-explanatory, particularly in highlighting digital literacy and financial barriers.

However, I would like to expand on your points and introduce some critical perspectives.

You suggest that a fully digital enterprise requires 100% digital literacy among employees. While this is an ideal form, research suggests it is not always realistic in all cases. Studies indicate that digital literacy varies due to generational differences, learning adaptability, and resistance to change (Järvinen and Karjaluoto, 2015).

A tiered training approach, tailoring cybersecurity education based on skill levels, might be a more sustainable alternative in the long run rather than expecting full digital proficiency.

You also mention multi-factor authentication (MFA) challenges, which is a key cybersecurity issue. However, beyond employee reluctance, poor usability of MFA systems can hinder security adoption. Research has shown that employees often circumvent security protocols when they perceive them as complex or disruptive, increasing vulnerability to cyber threats, which is linked to the human factor (Rosenberg et al., 2021).

Would you agree that usability improvements in MFA could significantly enhance cybersecurity adoption in SMEs, or do you believe more rigorous enforcement of security policies is necessary?

Regarding financial constraints in digital transformation, you correctly point out that SMEs struggle with funding. However, research suggests that beyond limited budgets, SMEs lack access to expert cybersecurity guidance, making them highly vulnerable (Williams et al., 2025) support in the form of government subsidies or public-private partnerships could help bridge this gap, but adoption rates remain slowish due to lack of awareness and bureaucratic complexity (Mavani et al., 2024).

Do you think that mandatory cybersecurity compliance standards would be more effective than voluntary financial incentives in encouraging SMEs to adopt better security measures?

Best wishes,
Franz

Through crafting my response, I believe that I developed my interpersonal skills, as my research led me to the conclusion that neither of Franz's suggestions regarding motivations to adopt security measures appeared to be correct. I therefore had to communicate this tactfully:

PEER RESPONSE

by [Todd Edge](#) - Friday, 7 March 2025, 4:57 PM

Hi Franz,

Many thanks for your insightful response. Your questions are thought-provoking and influenced me to research further into these issues!

Although I agree with your suggestion that poor usability of tools such as multi-factor authentication could be an influential factor to employee adoption of such digital processes, I also believe that my school's change management processes are currently insufficient, with goals and implications not fully communicated as per Ahmed et al.'s (2024) findings. By more clearly outlining the need for tools such as MFA and how staff members can better assist with safeguarding pupils by using this method, I believe that a smoother transition to adoption of such digital processes may be possible.

As to your question regarding whether mandatory cybersecurity compliance standards would be more effective than voluntary financial incentives in encouraging SMEs to adopt better security measures, this sounds like the age-old carrot vs stick debate! However, Chen, Chen & Wu (2018) found that neither perceived penalty nor reward had a significant impact on employees' compliance intentions; of much more significance were employee awareness of the importance of the security processes and their own capabilities to comply with standards (Chen, Chen & Wu, 2018). This emphasis on employee awareness seems to mirror Ahmed et al.'s (2024) findings that the adoption of digital processes is more comprehensive when employees are explicitly educated in the implications of said processes. In their respective studies, Ahmed et al. (2024) and Annisa, Siahaan & Lumbanraja (2024) also found employees' digital competence to be a significant factor in influencing organisational technological integration.

Thank you again for your questions; I enjoyed delving deeper into these issues!

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1. Identify and analyse security threats and vulnerabilities in network systems and determine appropriate methodologies, tools and techniques to manage and/or solve them.

Learning Outcome:

4. Articulate the legal, social, ethical and professional issues faced by information security and risk professionals.

By the time I approached the first assessment task, both my confidence and interest in the subject matter had improved and my desire to work autonomously to complete

the baseline assessment had definitely grown. This is consistent with Acosta-Gonzaga & Ramirez-Arellano's (2021) findings that students motivated by intrinsic factors are more likely to be able to engage in self-regulated learning.

Evidence of baseline assessment findings:

<u>Top-10 general web application security vulnerabilities (OWASP, 2021)</u>	<u>E-commerce/ JavaScript website-specific vulnerabilities</u>
Broken Access controls	Prototype pollution can be enacted in JavaScript-based websites which can lead to objects receiving values that they were not permitted to have (Kluban, Mannan & Youssef, 2024).
Cryptographic Failures	Ecommerce applications involve data transmission failure involving faulty digital signatures and encryption for security (Yu et al., 2023) – such encryption may not be secure within the analysed e-commerce website.
Injection	-Formjacking: malicious code injected into a web form to extract then send sensitive data to an attacker (Dharmavaram, 2021); -XML external entity injection attacks from misconfigured CML parsers (Shahid et al., 2022).
Insecure Design	An unsafe regular expression to sanitise input has been found within JSON.parse polyfill by some web domains – this can lead to a ReDoS attack (Kluban, Mannan & Youssef, 2024).
Security Misconfiguration	-In e-commerce websites, merchant systems incorporate various payment methods from TPPs (Yu et al., 2023) – this increases the attack surface and likelihood of security misconfigurations; -Lack of Content Security Policy (CSP) headers on the backend to tell the browser what behaviours are expected from client-side application (Freeman, 2024).
Vulnerable and Outdated Components	-Outdated JavaScript framework versions such as Angular JS can lead to prototype pollution attacks (Kluban, Mannan & Youssef, 2024).

Identification and Authentication Failures	-E-commerce systems rely on distributed and concurrent interactions among web applications involving clients, merchants, and third-party payment platforms (TPPs) for cashier services (Yu et al., 2023) – such processes could involve identification and authentication failures. - Merchants update the order status after the payment is completed (Yu et al., 2023) – this is another possible attack vector.
Software and Data Integrity Failures	- Session management: maintaining session states is crucial for security, where cookies can contain orderID and other sensitive data (Yu et al., 2023). Cookies containing such sensitive data could be compromised cross-site scripting (XSS) (Freeman, 2024). - Data validation: merchants use functions to validate the integrity of the transaction amount (Yu et al., 2023) – these could contain flaws that open up an attack vector.
Security Logging and Monitoring Failures	Ignored login failures, transaction failures, unclear error warnings (Armando & Rosalina, 2023).
Server-Side Request Forgery	Malicious users can potentially alter http control and data flows through concurrent interactions due to logical errors or a lack of server-side validations (Yu et al., 2023; Armando & Rosalina, 2023).

Learning Outcome:

3. Gather and synthesise information from multiple sources (including internet security alerts and warning sites) to aid in the systematic analysis of security breaches and issues.

Positive feedback from my tutor on the baseline assessment further boosted my perceived levels of competence:

Knowledge and understanding of the topic/issues under consideration (25%):

You have provided a good level of knowledge & understanding in relation to the vulnerability audit and assessment for your chosen website. You have also included a brief baseline/introduction into the type of website and identified a range of security challenges which are listed. You have indicated the stages you will go through, with some justification. You have also mentioned the frameworks and standards - GDPR etc. Further research has been used to support your analysis.

Distinction

Application of knowledge and understanding (25%):

There is fair demonstration of the application of knowledge with some mention of the tools, although this could have been more specific. You have listed potential security challenges and then you could have discussed specific tools/models required to address each of the issues. You have considered the impact to the business.

Merit

Criticality (25%):

There is a range of further research listed in the references list, and this is evidenced in the body of your writing. Your research focuses on the security vulnerabilities etc. You have considered the limitations. You have also provided a general timeline that the assessment process would take.

Merit

Structure and Presentation (25%):

A generally well organised report with the sections suitably laid out. References have been listed in the correct format. Spelling and grammar are good throughout.

Distinction

By researching and building up my knowledge of tools to be used, my motivation to use them seems to have increased, in line with Shapiro et al.'s (2017) findings. However, I still lacked confidence in performing manual testing as this appeared to require a greater level of understanding of backend website components. Tutor guidance within seminars however helped this to be a less arduous task, guiding my efforts towards confirming vulnerabilities that had been found.

Tutor guidance from seminar transcript:

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227
00:22:35.490 --> 00:22:41.690
Beran Necat: A manual testing is more like confirming the results of the automated scans.

228
00:22:41.970 --> 00:22:47.680
Beran Necat: Okay, which basically the tools that you used before

229
00:22:48.340 --> 00:22:57.560
Beran Necat: this could range from like, you know, end map nicta openvas. I'm sure, Nessus, there are so many tools out there.

230
00:22:58.400 --> 00:23:02.850
Beran Necat: and that way. You can actually start using

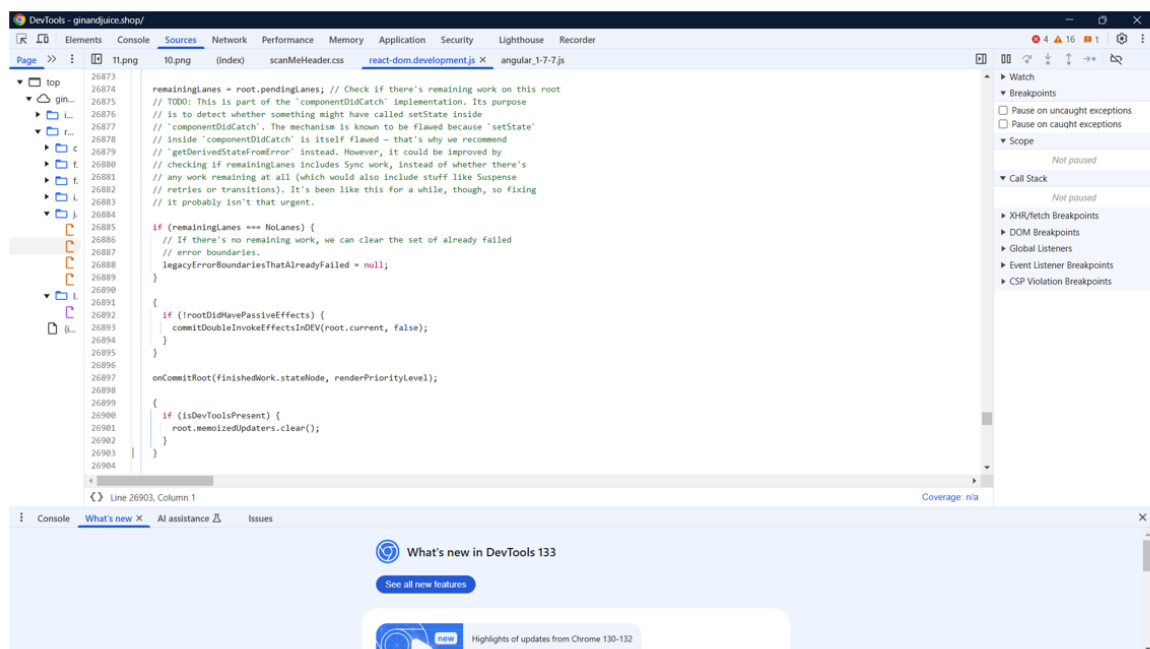
231
00:23:03.270 --> 00:23:17.680
Beran Necat: vulnerability, scanning through the automated tools things like SQL, injection, for example, which you can try to demonstrate this.

232
00:23:17.930 --> 00:23:25.380
Beran Necat: Okay? So if you are adopting this strategy, then we would like to see some output

233
00:23:25.940 --> 00:23:28.009
Beran Necat: based on SQL injection.
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This advice helped me to better manage my time during the manual testing process and I even began to enjoy this task, experiencing a sense of achievement, especially after finding evidence of the 'suspicious comments' vulnerability.

Evidence of suspicious comments confirmed by manual test:



Learning Outcome:

1. Identify and analyse security threats and vulnerabilities in network systems and determine appropriate methodologies, tools and techniques to manage and/or solve them.

Subsequent research into the revealed information regarding the React component and inferences made regarding its threat to the system:

The Information Disclosure - Suspicious Comments vulnerability would fall under the 'Insecure Design' header from the baseline assessment; the comments refer to a feature of a software component used to construct many of the webpages ('React'). Within the comments (see Appendix 2), lines of code are discussed that reveal where software errors will not be handled (Facebook Open Source, 2025) – potentially allowing an attacker to crash the software, effectively causing a Denial of Service (DOS) attack.

Learning Outcome:

1. Identify and analyse security threats and vulnerabilities in network systems and determine appropriate methodologies, tools and techniques to manage and/or solve them.

Learning Outcome:

2. Design and critically appraise computer programs and systems to produce solutions that help manage and audit risk and security issues.

What next?

I have come to realise that much of the knowledge gained from this module can be directly used in my role as an IT Technician: the research into digital transformation and subsequent security adoption can inform strategies to help me to better facilitate the uptake of more secure digital practices. Furthermore, I have very recently been asked to develop a new website for my school and so my new-found abilities to conduct and interpret vulnerability analyses has become an important skill to utilise.

In terms of my future studies, I have learnt to have more belief in my ability to quickly build on knowledge, and I now understand better that knowledge breeds motivation (Shapiro et al., 2017). Ultimately, I have learnt that I should aim to keep an open mind about new modules, not writing them off as being 'not my area' and not writing myself off if I feel that I initially have less knowledge than others.

References

Acosta-Gonzaga, E., & Ramirez-Arellano, A. (2021) The Influence of Motivation, Emotions, Cognition, and Metacognition on Students' Learning Performance: A Comparative Study in Higher Education in Blended and Traditional Contexts. *SAGE Open* 11(2): 1-12. DOI: 10.1177/2158244021102756

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